

Evaluating the performance of large language models in drafting data management plans

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Source Code: <https://github.com/fairdataihub/nih-dmp-llm-generation>,
<https://github.com/fairdataihub/nih-dmp-llm-evaluation-paper-code>

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Dataset:

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Main Text of Abstract

Most funders require Data Management Plans (DMPs) to be submitted as part of a research proposal. A DMP is a document that describes how researchers will manage, preserve, and share data from their projects in line with the Findable, Accessible, Interoperable, and Reusable (FAIR) principles and other relevant disciplinary guidelines. Their aim is to ensure researchers plan in advance for effective data management and sharing. Preparing a DMP is usually difficult for researchers because they often lack training and knowledge in data management practices. Moreover, funding organizations have different policies that require adherence to specific data management strategies and DMP formats. For instance, the United States National Institutes of Health (NIH) and the US National Science Foundation (NSF) both fund biomedical research but differ in their DMP requirements. Librarians are available to assist at larger institutions, but they face similar challenges and do not always have the resources to assist with all aspects of the plan. Open source and free platforms, such as DMPTool, are available to support researchers by providing funder-specific guidance, sample DMPs, and a convenient user interface for creating compliant DMPs. Yet it is still up to researchers and librarians to complete the various sections of their DMPs, which vary widely depending on factors such as the research domain and funder requirements.

In this work, we evaluated the performance of Large Language Models (LLMs) in drafting DMPs compliant with the NIH's 2023 guidelines. The goal was to investigate whether LLMs were ready, off-the-shelf, to assist biomedical researchers with preparing their DMPs. We evaluated the off-the-shelf performance of an open-source LLM, Llama 3.3, and a commercial LLM, GPT-4.1, in drafting NIH-compliant DMPs. We tested the models' performance in generating 26 human-written DMP examples provided by the NIH from previously funded proposals across different biomedical research domains and data types. According to the 2023 NIH guidelines, each DMP

follows a standard structure consisting of 12 sections. Each section requires information on various aspects of data management and sharing, including which data types will be collected, the standards for data and metadata formatting, where and how data will be shared, and more. Our prompting strategy consisted of a single prompt that included the name of the NIH Institute/Center to which the proposal was submitted, details about the data collection (whether human or non-human participants, number of subjects, and data types to be collected), and the full NIH-provided DMP template. We evaluated the generated DMP using two evaluation approaches: automatic reference-based evaluation and human expert review. To compare the 12 sections of the LLM-generated DMPs with NIH samples, we used two metrics to compute semantic and textual similarity: ROUGE (Recall-Oriented Understudy for Gisting Evaluation) and SBERT (Sentence-Bidirectional Encoder Representations from Transformers). For the human expert evaluation, we developed a custom survey that guided participants step-by-step in evaluating the three versions of different NIH DMPs (human-written, Llama-generated, and GPT-generated). The questions included how satisfied they were with each element and what type of errors they found. At the end, they provided an overall evaluation of each DMP and indicated whether they believed it was LLM-generated or human-written.

The automatic reference-based evaluation scores for both ROUGE and SBERT were slightly higher for GPT (ROUGE score in the range of 0.55 to 0.71 and section-wise SBERT score in the range of 0.19-0.38) than for Llama (0.53-0.67 and 0.18-0.36) when we compared them to the NIH-provided samples. In particular, GPT achieved higher similarity to NIH references than Llama in sections tied to repository selection, access conditions, reuse considerations, and oversight. A total of 21 human experts participated in the evaluation. On a 5-point Likert satisfaction scale, 90.5% of the experts (19 out of 21) reported being satisfied or very satisfied with the DMPs generated by GPT. In contrast, 61.9% of the experts (13 out of 21) reported they were dissatisfied or very dissatisfied with the DMPs generated by Llama. GPT outputs were most often mistaken for human-written text, with only 14.3% (3 out of 21) correctly identifying them as LLM-generated. Overall, GPT showed the lowest error rates across error types in the assessment.

Overall, our results suggest that LLMs can accelerate the creation of initial drafts, but expert review remains essential before submission to ensure full correctness and compliance. Although GPT received higher ratings in both automated and human evaluations, it still needs to be tailored to the specific study and the research team's goals. Moreover, open-source LLMs may not be as ready off-the-shelf as commercial ones for drafting funding-compliant DMPs. In this presentation, we will introduce our motivation for this work, present our methods, and discuss our results. We believe this presentation will be valuable to the BOSC community, as many participants likely have to navigate DMP requirements for their funding proposals. We look forward to discussing how open-source LLMs can be further developed and adapted to better support DMP requirements and FAIR data practices more broadly in biomedical research.